



Deep sea decapod crustaceans of São Pedro and São Paulo Archipelago, Equatorial Atlantic, Brazil

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Abstract

Six species of deep sea decapod crustaceans (*Chaceon gordonae*, *Bathynectes longispina*, *Dromia bollorei*, *Homola minima*, *Nematocarcinus gracilipes* and *Plesionika martia*) were sampled around the vicinity of São Pedro and São Paulo Archipelago, Brazil (0°55'N; 29°21'W), after nine scientific surveys to explore demersal and benthic species in the area. Specimens were collected by bottom traps between 200 and 700 m deep, from 2012 to 2015. The current record enlarges the knowledge of the distribution of five of these species, new to the Archipelago. *Chaceon gordonae* was previously known to the area.

Key words: Brazilian Oceanic Island, Mid-Atlantic Ridge, Brachyura, Caridea, *Chaceon gordonae*

Resumo

Seis espécies de crustáceos decápodes de águas profundas (*Chaceon gordonae*, *Bathynectes longispina*, *Dromia bollorei*, *Homola minima*, *Nematocarcinus gracilipes*, e *Plesionika martia*) foram capturadas durante expedições científicas realizadas no entorno do Arquipélago de São Pedro e São Paulo, Brasil (0°55'N; 29°21'W). Os espécimes foram coletados por armadilhas de fundo em profundidades entre 200 e 700 m, entre 2012 e 2015. O registro atual aumenta o conhecimento sobre a distribuição de cinco dessas espécies, novas para o Arquipélago. *C. gordonae* já havia sido registrada previamente para a área.

Introduction

The São Pedro e São Paulo Archipelago (SPSPA) is a small and remote group of islands, located in the Equatorial Atlantic Ocean, on the Mid-Atlantic Ridge (0°55'10"N e 29°20'33"W), between the African and American continents and distant 1010 km off the Brazilian coast (Lubbock & Edwards, 1981). The SPSPA is governed by a complex system of currents (Molinari, 1982; Stramma, 1991) and is under the direct influence of the Intertropical Convergence Zone (ITCZ) (Travassos *et al.*, 1999). The resulting oceanographic and atmospheric conditions directly affect the life cycle of different taxonomic groups (Leite *et al.*, 2008; Lima *et al.*, 2011; Tibúrcio *et al.*, 2011; Macedo-Soares *et al.*, 2012; Melo *et al.*, 2012; Vieira *et al.*, 2012; Branco *et al.*, 2013).

Studies dealing with crustaceans in the vicinity SPSPA mainly focused on the description of shallow water species (Houltuis *et al.*, 1980; Viana *et al.*, 2004), larvae of decapod (Brandão *et al.*, 2013), aspects of the biology

of the crab *Grapsus grapsus* (Freire *et al.*, 2010) and of the lobster *Panulirus echinatus* (Pinheiro *et al.*, 2003). Indeed, few studies have been conducted so far to explore deep sea species of crustaceans in oceanic Brazilian islands. Most of the information available is restricted to the mid- and upper continental slope and seamounts and was obtained during the REVIZEE Program (Program for the Assessment of the Sustainable Yield of Living Resources of the Exclusive Economic Zone) (Silva *et al.*, 1998; Viana *et al.*, 2003; Coelho Filho, 2006; Pezzuto *et al.*, 2006; Coelho *et al.*, 2007; Serejo *et al.*, 2007; Souza-Filho & Serejo, 2008; Arana *et al.*, 2009; Cardoso *et al.*, 2014; Oliveira *et al.*, 2014; Ferreira *et al.*, 2016). Despite such studies, the knowledge on deep sea crustaceans associated to the oceanic islands, such as the SPSPA, is still fragmentary in Brazil.

Material and methods

Since 2012, scientific expeditions to explore the deep waters at SPSPA have reported new occurrences and new species, at depths of more than 200 m (Carvalho-Filho *et al.*, 2016; Ferreira *et al.*, 2016; Nunes *et al.*, 2016), including crustaceans taken by baited traps around the archipelago at depths between 200 and 700 m (Figure 1). Position (latitude, longitude) and depth (m) of each trap were always recorded, as well as the sex and number of the specimens caught. All specimens were labelled, preserved at ethanol 70% and brought to laboratory for detailed studies. Some were deposited at “Museu de Oceanografia Prof. Petrônio Alves Coelho” (MOUFPE) University of Pernambuco.

At the laboratory, specimens were measured using a vernier caliper (0.01 mm), including carapace width (CW: the distance between the rear bases of the fifth pair of anterolateral teeth), for Brachyura specimens, and carapace length (CL: measured from the groove between the first pair of front teeth to the posterior margin of the carapace), for Caridea specimens. Individuals were identified according to the available literature (Crosnier & Forest, 1973; Forest, 1974; Takeda & Okutani, 1983; Ingle, 1985; Manning & Holthuis, 1981; Guinot & Richer de Forges, 1995; Tavares, 2003; Tavares & Pinheiro, 2011; Cardoso & Burukovsky, 2014; Tavares & Lemaitre, 2014). The classification adopted was based on Martin & Davis (2001).

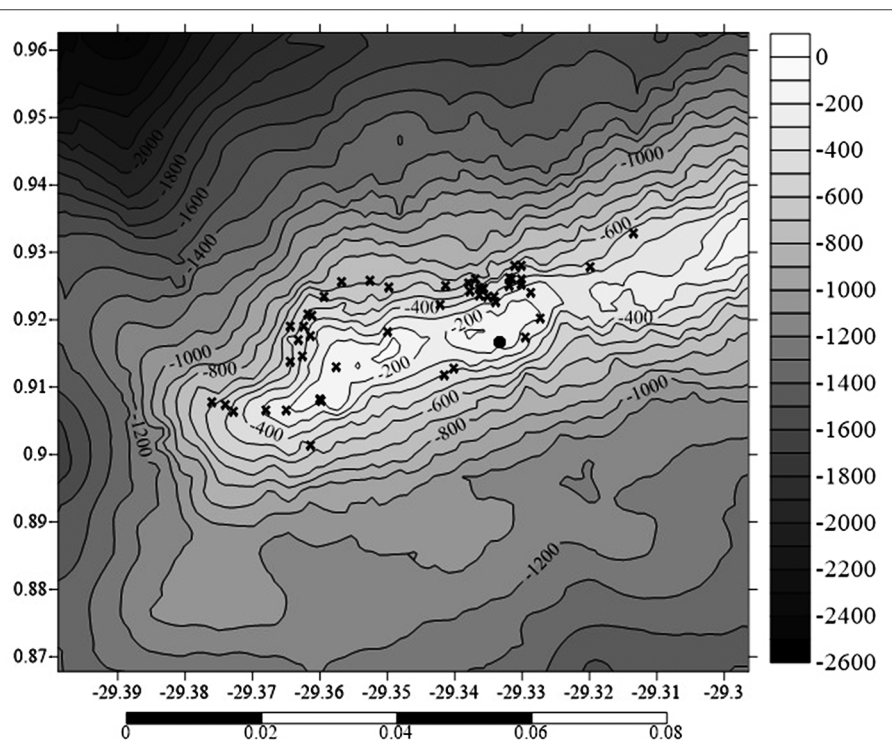


FIGURE 1. Location of surveys with bottom traps for catch deep sea crustacean species in the vicinity of São Pedro e São Paulo Archipelago, in depth ranges of the 200-700 m (x=stations; ●=SPSPA).

Results

General aspects. Six distinct species of deep sea crustaceans were collected around SPSPA, between 2012 and 2015: *Chaceon gordonae* (Ingle, 1985), *Bathynectes longispina* (Stimpson, 1871), *Dromia bollorei* (Forest, 1974), *Homola minima* (Guinot & Richer de Forges, 1995), *Nematocarcinus gracilipes* (Filhol, 1884), and *Plesionika martia* (A. Milne-Edwards, 1883). The dominant species with wide bathymetric distribution was the Geryonidae crab *C. gordonae* (93.5 %), followed by the other species, all of them with a much smaller number of specimens collected (Table 1).

TABLE 1. Species collected in number per depth strata during scientific surveys in the vicinity of São Pedro e São Paulo Archipelago, Equatorial Atlantic, Brazil.

Species	Depth Strata (m)						Total
	100–200	200–300	300–400	400–500	500–600	600–700	
<i>Chaceon gordonae</i>	0	18	167	149	301	66	701
<i>Bathynectes longispina</i>	8	1	5	1	0	0	15
<i>Dromia bollorei</i>	0	0	0	1	8	0	9
<i>Homola minima</i>	0	0	0	1	0	0	1
<i>Plesionika martia</i>	0	0	1	0	0	0	1
<i>Nematocarcinus gracilipes</i>	0	0	1	7	14	1	23
Total	8	19	174	159	323	67	750

Order Decapoda

Suborder Pleocyemata Burkenroad, 1963

Infraorder Brachyura Latreille, 1803

Section Eubrachyura de Saint Laurent, 1980

Subsection Heterotremata Guinot, 1977

Superfamily Portunoidea Rafinesque, 1815

Family Geryonidae Colosi, 1923

Genus *Chaceon* Manning & Holthuis, 1989

Chaceon gordonae (Ingle, 1985)

(Figure 2)

Geryon gordonae Ingle, 1985: 90, fig.: 1, 2, 5a.

Chaceon gordonae—Manning and Holthuis, 1989: 8.—Afonso-Dias *et al.*, 2008: 1, fig. 2a.

Material examined. 2♀, MOUFPE 15256, São Pedro e São Paulo Archipelago (0°54'787"N e 29°21'451"W), 200 m, June 2012. 2♂, MOUFPE 15257, São Pedro e São Paulo Archipelago (0°54'787"N e 29°21'451"W), 700 m, December 2013 (Table 2).

Diagnosis. Lateral part of orbital dorsal margin slightly concave, first anterolateral tooth produced, inner infra-orbital tooth/antennal third peduncular segment noticeably depressed, dactylus of pereopods 2–5 laterally compressed, submedian frontal teeth on a frontal projection, inner orbital teeth spinosed, breadth of orbit (measured from inner to outer orbital tooth) noticeably less than width of frontal margin. Upper margin of cheliped palm with distal tooth, outer anterior margin of carpus with a tooth in small males. Dorsal margin of merus with

distal and subdistal teeth. Merus of pereopods 4–5 with antero-dorsal tooth, third maxilliped merus with length/breadth as broad as long, third maxilliped merus with outer margin disto-externally angular and, finally, carapace broader than long (Ingle, 1985; Afonso-Dias *et al.*, 2008).

Distribution. Geryonid crabs are widely distributed in the world's ocean, commonly occurring on continental slopes at depths of 100–2,000 m (Manning & Houlthuis, 1989). Ingle (1985), based on two males and two females caught at 1,153 m, from southern Iceland to Sierra Leone Ridge, off western Africa, reported the occurrence of *Chaceon gordonae*, in the Atlantic Ocean.

Afonso-Dias *et al.* (2008), analyzing collections of São Tomé e Príncipe Archipelago, also off western Africa, was the second to report the specie in the North Atlantic Ocean. They found *C. gordonae* in reasonable quantities, enough to sustain a small artisanal fishery since 1980. The species was reported for the first time from SPSPA, by Ferreira *et al.* (2016).

TABLE 2. Measurements of *Chaceon gordonae* collected in the São Pedro e São Paulo Archipelago, Equatorial Atlantic, Brazil

Specimens	Parameters						
	CW	CL	LQL	LQW	RQL	RQW	AW
1	79.1	56.7	41.1	13.3	42.6	13.9	25.2
2	75.1	48.5	38.7	12.7	36.9	11.4	21.1
3	122.1	88.3	72.1	22	76.6	24.6	-
4	130	106	92	32	94	35	-

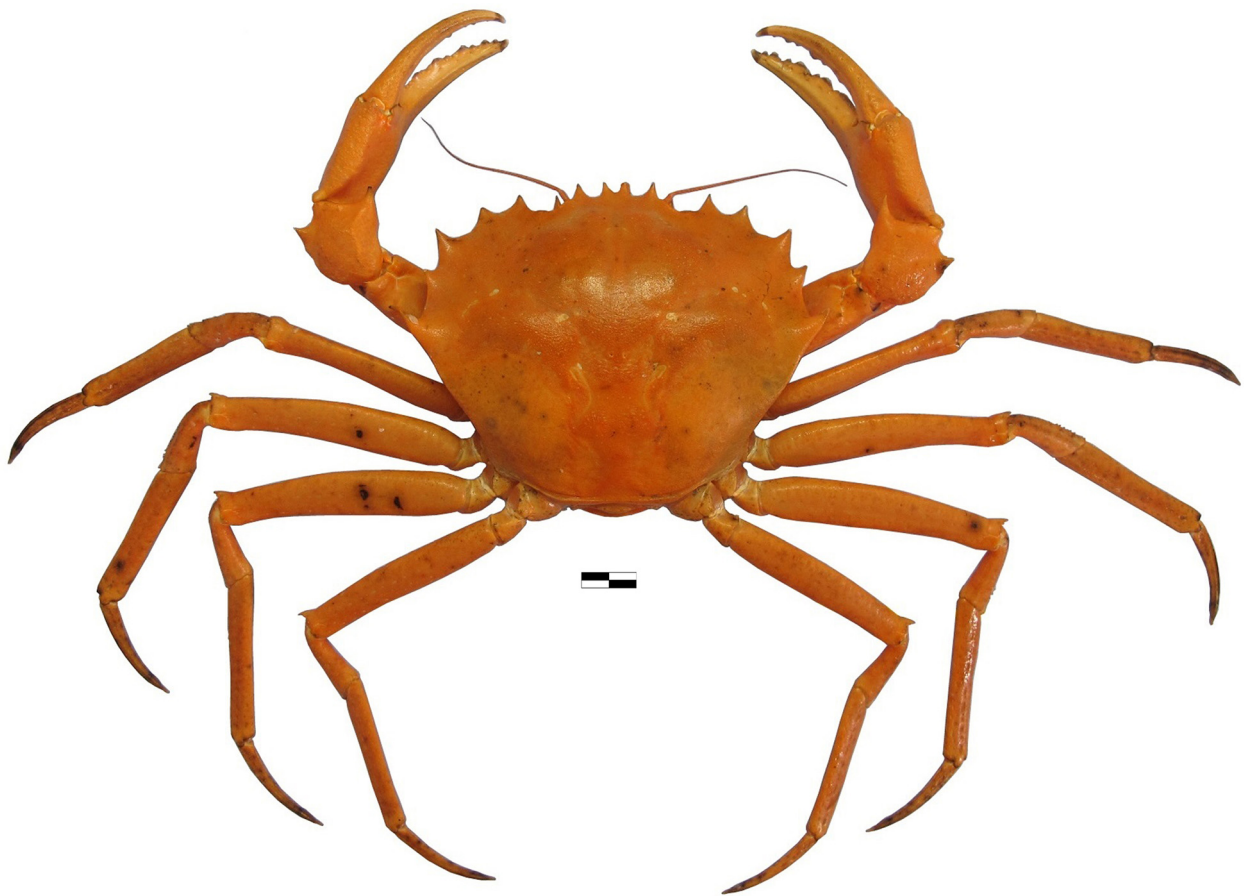


FIGURE 2. *Chaceon gordonae* (Ingle, 1985), collected off São Pedro e São Paulo Archipelago, Brazil, MOUFPE 15256/15257. Scale bars 10 mm

Remarks. The Geryonidae family currently has three genera: *Geryon* with five species, *Chaceon* with 28 species and *Zariquieyon* with a single species (Manning & Holthuis, 1989; Ng *et al.*, 2008). In Brazil, there were three species of the genus *Chaceon* already reported: *C. notialis* Manning and Holthuis, 1989; *C. ramosae* Manning, Tavares and Albuquerque, 1989; and *C. linsi* Tavares & Pinheiro, 2011. *Chaceon gordonae*, collected at SPSPA, represents the fourth species occurring in Brazilian waters (Pezzuto *et al.*, 2006; Tavares & Pinheiro, 2011; Ferreira *et al.*, 2016).

Family Portunidae Rafinesque, 1815

Subfamily Polybiinae Ortmann, 1893

Genus *Bathynectes* Stimpson, 1871

Bathynectes longispina Stimpson, 1871

(Figure 3)

Bathynectes longispina Stimpson, 1871:146.—Tavares, 2003: 1, fig. 1-2.—Torres *et al.*, 2006: 134, fig. 1.

Material examined. Eight specimens (Table 3; 6♀/2♂), MOUFPE 15259, São Pedro e São Paulo Archipelago (0°54'787"N e 29°21'451"W), 420 and 546 m, April 2014.

Diagnosis. Carapace with five anterolateral teeth, the last one twice as long as the others; four frontal teeth. Right cheliped larger than the left one, both with carinae and spines (Tavares, 2003; Torres *et al.*, 2006).

Distribution. *Bathynectes longispina* was previously reported for the east coast of North America, Gulf of Mexico and Cuba (Powers, 1977; Manning & Holthuis, 1981; Ingle, 1985). It was later also found in Brazilian waters (Tavares, 2003; Torres *et al.*, 2006). This was the first time the species was collected at SPSPA representing a new record for the area.



FIGURE 3. *Bathynectes longispina* (Stimpson, 1881), collected off São Pedro e São Paulo Archipelago, Brazil, MOUFPE 15259. Scale bars 10 mm

Remarks. The specimens examined herein fit well with the descriptions provided by Stimpson (1871), Rathbun (1930), the later as *B. superba*, and Tavares (2003). As pointed out by Tavares (2003), our material is similar to the material collected from Espírito Santo state, which has the chelipeds unequal, the right one being larger than the left one.

A total of 15 specimens were collected. Eight were analyzed, being two males and six females, with carapace lengths ranging from 54.1 to 69.4 mm CL and width from 75.8 to 99.5 mm CW. For females the width of the abdomen ranged from 41 to 44.6 mm.

TABLE 3. Measurements of *Bathynectes longispina* collected in the São Pedro e São Paulo Archipelago, Equatorial Atlantic, Brazil

Specimens	Parameters		
	CW	CL	AW
1	94.2	67.7	-
2	90.8	67.0	43.9
3	97.6	68.0	44.6
4	99.5	69.4	44.4
5	89.5	62.2	41.0
6	93.0	66.2	44.1
7	78.8	54.1	-
8	90.7	60.8	42.6

Superfamily Dromioidea De Haan, 1833

Family Dromiidae De Haan, 1833

Subfamily Dromiinae De Haan, 1833

Genus *Dromia* Weber, 1795

Dromia bollorei Forest, 1974

(Figure 4)

Dromia bollorei Forest, 1974: 91, fig. 1d, 2, 3d, 5, 6b, 7c, d; pl. 2, fig. 1,2; pl. 3, fig. 4; pl. (5, lig. 1).—Maning & Holthuis, 1981: 11.—McLay, 1993: 151 (key).—Guinot & Tavares, 2003: 94: fig. 27b.—Cleva *et al.*, 2007: 240, fig. 8c.—Ng *et al.*, 2008: 33.

Material examined. Nine specimens (Table 4; 6♀/3♂), MOUFPE 15258, São Pedro e São Paulo Archipelago (0°54'787"N e 29°21'451"W), 420 and 546 m, December 2013.

Diagnosis. Carapace and pereopods pilose. Carapace front with two strong triangular teeth apically blunt; mesobranchial region with two humps, the innermost being particularly prominent. Anterolateral edge with four strong teeth, three of which are regularly conic: the first and the third slightly larger and the fourth shorter and wider than the previous ones, with a very lateral orientation; the second tooth is also conical, with its posterior margin with obsolete projection, corresponding to the accessory tooth. Chelipeds relatively strong and very elongated in large males specimens, with merus exceeding at least half of the length of the anterolateral edge of the carapace. Palm elongated, with its height in the middle portion ranging from 2.6 to 3 times total length; fixed finger strongly deflected in males, weaker in females. Second and third pereopod rather long and slender. Pereopod 2 propodus with ratio of the length and height close to 3. Pereopods 4 and 5 short, especially in large males. Pereopod 5 folded laterally on the shell, and far from reaching the tooth. Female sternal grooves end together between sternite 5.



FIGURE 4. *Dromia bollorei* (Forest, 1974), female collected off São Pedro e São Paulo Archipelago, Brazil, MOUFPE 15258. Scale bars 10 mm

TABLE 4. Measurements of *Dromia bollorei* collected in the São Pedro e São Paulo Archipelago, Equatorial Atlantic, Brazil

Specimens	Parameters						
	CW	CL	LQL	LQW	RQL	RQW	AW
1	71.0	60.5	35.8	13.4	35.2	13.8	35.8
2	68.4	57.1	32.1	13.2	-	-	32.5
3	83.7	68.5	68.9	30.1	66.5	29.6	-
4	73.7	62	35.9	13.8	35.3	13.9	35.6
5	64	54.2	38.5	16.9	36.9	16.4	-
6	76.9	63.1	37.3	13.8	35	13.4	37
7	69.9	58.2	34.2	13.5	33.9	13.9	34.1
8	76.9	63.1	37.5	13.8	37.9	14.2	36.1
9	63.1	51.6	35	14.1	34.2	14.9	-

Distribution. Known only from Mauritania and Ivory Coast, at 100 m (Manning & Holthuis, 1989). This is the first record of this species for the São Pedro e São Paulo Archipelago, increasing their range in the equatorial region.

Remarks. The specimens analyzed in this study showed the main diagnostic characters described by Forrest (1974), such as: frontal teeth triangular and strong, mesobranchial region with two humps, first, second and fourth

anterolateral teeth conic and strong, the third and females with sternal grooves which end together between sternite 5 (figure 5). However, comparing our material with photographs of the types series, holotype female - MNHN-IU-2008-11166, paratypes: 1 female MNHN-IU-2008-11162, 1 male MNHN-IU-2008-11163, (available on line at <http://coldb.mnhn.fr/catalognumber/mnhn/iu/2008-11166>), the lateral spines of carapace are less acute.



FIGURE 5. Urogenital view of *Dromia bollorei* (Forest, 1974), collected off São Pedro e São Paulo Archipelago, Brazil. Scale bars 10 mm

Nine individuals were collected, three of them being male and six female. The carapace length ranged from 51.6 to 68.5 mm, and the width from 63.1 to 83.7 mm. All specimens were covered with marine sponges, a characteristic of the species, which is, for this reason, known as "sponge crab".

Superfamily Homoloidea De Haan, 1839

Family Homolidae De Haan, 1839

Genus *Homola* Leach, 1815

Homola minima Guinot & Richer de Forges, 1995

(Figure 6)

Thelxiope barbata—Rathbun, 1937: 63, fig. 16, tbl. 17, pl. 15, figs. 1,2;—Chace, 1940: 8.

Homola barbata—Williams, 1984: 261, fig. 193;—Melo, 1985: 32; 1996: 75, unnumbered fig; 1999: 440, fig. 2;—Melo *et al.*, 1998: 440;—Nizinski, 2003: 123. [not *Homola barbata* (Fabricius, 1793)]

Homola minima Guinot and Richer de Forges, 1995: 326, figs. 8c,d, f, 9b.;—Martin & Zimmerman, 2001: 412; Nizinski, 2003: 123;—Felder *et al.*, 2009b: 1072;—Tavares & Lemaitre, 2014: 514, figs. 6, 7, 8, 9.

Material examined. One specimen with 47,3 mm CL e 38,5 mm CW, male, MOUFPE 15261, São Pedro e São Paulo Archipelago (0°55'681"N e 29°19'804"W), 470 m, 12/29/2013.

Diagnosis. Small species, carapace urn-shaped; rostrum bifurcate at tip, one spine on each side at base of rostrum, one at outer orbital angle; eyestalks long, slender at base. Cheliped surface granulated and hairy. Pereopod 5 folded forward over back, yellowish-brown or reddish-brown hairs, spines red; merus with distal end reaching the cervical groove.



FIGURE 6. *Homola minima* Guinot & Richer de Forges, 1995, collected off São Pedro e São Paulo Archipelago, Brazil, MOUFPE 15261. Scale bars 10 mm

Distribution. *Homola minima* is distributed in the Western Atlantic, from Massachusetts to Florida in the Northeastern coast of the United States, including Bahamas, Gulf of Mexico, the Caribbean Sea, Suriname, Brazil and Uruguay. It occurs in depths ranging between 55 and 700 m. (Tavares & Lemaitre, 2014; Guinot & Richer de Forges, 1995). This is the first record of this to the São Pedro e São Paulo Archipelago.

Remarks. *Homola minima* Guinot & Richer de Forges, 1995 is very similar to *Homola barbata* (Fabricius, 1793). According to Tavares and Lemaitre (2014), both species show a great ontogenetic variation in the diagnostic characters originally used by Guinot & Richer de Forges (1995). Indeed, the only character that can be used to distinguish these species is the distal end of the merus of the pereopod 5, which reaches the cervical groove, at maximum, whereas it is distinctly over the cervical groove in *H. barbata*.

Infraorder Caridea Dana, 1852

Superfamily Nematocarcinoidea Smith, 1884

Family Nematocarcinidae Smith, 1884

Genus *Nematocarcinus* A. Milne-Edwards, 1881

Nematocarcinus gralicipes Filhol, 1884

(Figure 7)

Nematocarcinus gralicipes—Filhol, 1884: 232, fig. 1.—Crosnier & Forest, 1973:123, fig. 34-35.—Neves, 1982: 1, fig. 1-2.—

Burukovsky, 2009: 83, fig. 1.—Cardoso & Burukovsky, 2014: 440, fig. 2-4.

Nematocarcinus agassizii Faxon, 1893: 204.—Cardoso & Burukovsky, 2014: 441.

Material examined. 14 specimens, MOUFPE 15260, São Pedro e São Paulo Archipelago (0°55'681"N e 29°19'804"W), 700 m, 12/29/2013.

Diagnosis. Body moderately slender; integument moderately hard, not membranous, surface smooth, shiny. Rostrum proximally horizontal, distally it slightly rises up; rostrum length 70–96%, in the extreme case up to 120%, of carapace length; rostrum dorsal margin (including postrostral crest) with 19–36 (usually 23–32) tightly fitting teeth; rostrum ventral margin with 1–5, usually 3–4 teeth, lying in the region of the rostrum free of dorsal teeth. Eyes normally developed, its width noticeably larger than the diameter of the eye stalk. Posterodorsal margin of third abdominal somite poorly developed. Continuation of its sides when crossing forms an obtuse angle close to 90°. Fifth abdominal somite pleura with its sides intersecting at an angle with less than 90°; armed with a little spine, which axis coincides or nearly coincides with the bisector of the angle in adult specimens (carapace length of 15mm or more), in small shrimps the lower margin of the pleura is rounded and the spine is absent. Tubercles, rollers and other elevation on the inner surface of the anterior margin of the pleura of the fifth abdominal somite are absent. Distoventral organ at sixth abdominal somite formed by two single parallel rows of setae that begins with long plumose setae and at the proximal third of sixth abdominal somite length turns to short spindle shaped setae (not plumose) that extends to more than a half of the length of spots; spots located at the sideways surface of sixth abdominal somite; spots more or less oval or teardrop shaped; spots length not more than 1.5 times its width, distance between them 2–2.5 times its width. Telson with 6–9 (usually 7–8) pairs of dorsolateral spines; usually with one accessory spine, occasionally specimens with two or without accessory spines can be found (modified from Cardoso & Burukovsky, 2014).

Distribution. According to Burukovsky (2009), *N. gracilipes* has a distribution similar to many species that occur in the Portuguese coast and the Mediterranean Sea. It is common in the slope of the Moroccan coast (Burukovsky, 2012) and the Azores Archipelago, Cape Verde Archipelago and around Ascension (Crosnier & Forest, 1973). The occurrence in the Mid-Atlantic Ridge north of the Azores (Burukovsky, 2009) and South of the Equator, including the Brazilian coast (Cardoso & Burukovsky, 2014), confirms that the species prefers areas of seamounts. This is the first record of this species for the SPSPA, expanding its range of occurrence to the equatorial region of the Atlantic.

Remarks. The Nematocarcinidae family has five genera (*Lenzicarcinus*, *Lipkius*, *Nematocarcinus*,

Nigmatullinus, *Segonzackomaius*), with *Nematocarcinus* being the most numerous, with 48 species, among which stands out *N. gracilipes* (De Grave & Fransen, 2011). *Nematocarcinus gracilipes* differs from other species of the genus mainly by having: the rostrum in mature specimens overreaching antennular peduncle with more than one third of its length and reaching scaphocerite distal margin or slightly overreaching it; distoventral organ at sixth abdominal somite formed by spots 1.5–2 times longer than wide; and by single parallel rows of long plumose setae that are replaced by short spindle shaped setae (not plumose) at the proximal third and sixth abdominal somite, extending to more than a half of spots length (Cardoso & Burukovsky, 2014).

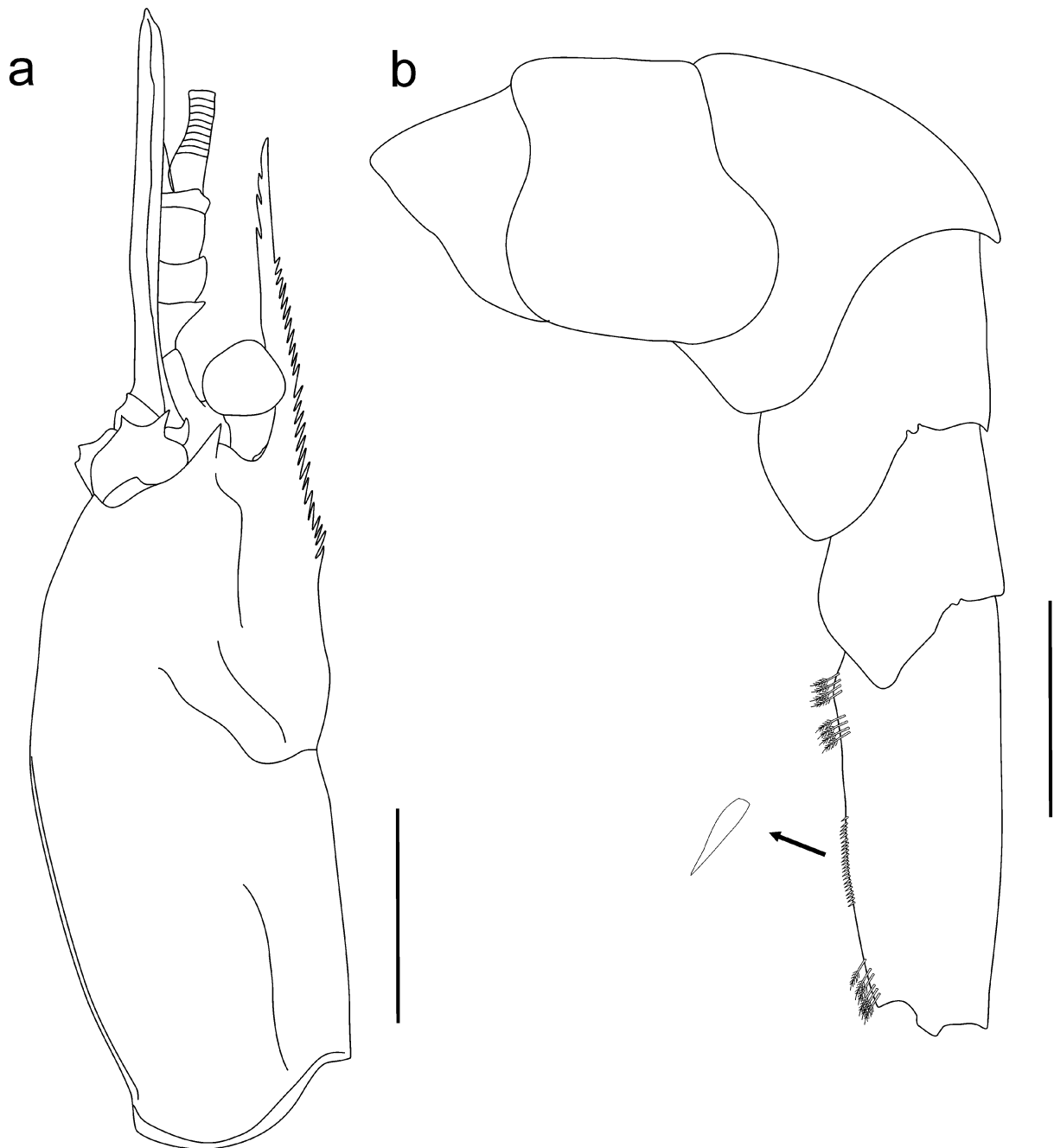


FIGURE 7. *Nematocarcinus gracilipes*, female (14.89 mm CC) collected off São Pedro e São Paulo Archipelago, Brazil, MOUFPE 15260. a) Lateral view of carapace; b) lateral view of abdominal somites 1–6. Scale bars 5 mm.

Superfamily Pandaloidea Haworth, 1825

Family Pandalidae Haworth, 1825

Genus *Plesionika* Bate, 1988

***Plesionika martia* (A. Milne-Edwards, 1883)**

(Figure 8)

Pandalus martius A. Milne-Edwards, 188, pl 21.

Plesionika martia—Crosnier and Forest, 1973: 112, figs. 63d, 64, 66 (see for extensive synonym).

Material examined. One specimen (TL 114.8 mm, CL 31.6 mm), MOUFPE 17899, São Pedro e São Paulo Archipelago (0°54.560' N e 29°21.895' W), 340 m, 25 August 2015.

Diagnosis. Carapace without lateral carinae; rostrum smooth dorsally for most of its length with few teeth (6–9 dorsal rostral teeth) at base, none movable. Posterior margin of third abdominal somite straight and not armed; no subapical tooth. Second maxilliped with terminal segment broader than long, applied as strip to distal margin of penultimate segment. Pereopod 2 equal in length. Strap-like epipods present in at least 2 anterior pairs of pereopods. Pereopod 4 with epipod. Telson longer than sixth abdominal somite, with 3 pairs of dorsolateral spines, including pair adjacent to lateral pair of posterior spines (Ahamed *et al.*, 2017).

Distribution. West Atlantic: off South Carolina to Florida and off Bermuda; NE and SW Gulf of Mexico. Eastern Atlantic, Azores: off SW Ireland, Bay of Biscay, throughout Mediterranean, Gulf of Guinea, and Cape of Good Hope. Asian waters: Andaman Sea, Bay of Bengal, Indian Ocean and Arabian Sea; at depths of 180–400 m (Pequegnat, 1970; Chace, 1985; Ahamed *et al.*, 2017).

Remarks. Chace (1985) states that the proportional size of the rostrum cannot be used to distinguish *P. martia* from other species such as De Man (1920) and others have attempted to do. In this sense, the size and shape of the eye (orbit configuration) appear to be the characteristic that differ from *P. martia* of the Indo-Pacific species (e.g. *Plesionika orientalis*). Another important feature is the length and proportion of the distal segments of the three posterior pairs of pereopods. However, these appendages are extremely fragile and are usually absent in the samples (Chace, 1985; Ahamed *et al.*, 2017).



FIGURE 8. *Plesionika martia*, female with TL 114.8 mm and CL 31.6 mm, collected off São Pedro and São Paulo Archipelago, Brazil, MOUFPE 17899. Scale bars 10 mm

Discussion

Six deep-sea decapod crustaceans are listed on this study, and five of them are new records for the SPSPA. These new findings represent an increase from 17 to 23 species recorded in the area (Holthuis *et al.*, 1980; Lubbock & Edwards, 1981; Sankarankutty *et al.*, 2001; Viana *et al.*, 2004) (Table 5).

TABLE 5. Detailed distributions of shallow and deepwater decapod crustaceans registered in the São Pedro and São Paulo archipelago (SPSPA), Equatorial Atlantic, Brazil.

Species	Regions					
	Caribbean	SPSPA	Ascension	St Helena	Cape Verde	West African
Shallow Waters						
<i>Domecia acanthophora</i> (Desbonne, 1867)	+	+			+	
<i>Euryozius sanguineus</i> Linnaeus, 1771		+	+	+		
<i>Grapsus grapsus</i> (Linnaeus, 1758)		+				+
<i>Lucifer faxoni</i> Borradaile, 1915	+	+				+
<i>Lucifer typus</i> H. Milne Edwards, 1837	+	+				+
<i>Lysmata amboinensis</i> (de Man, 1888)		+				
<i>Mithraculus forceps</i> (A. Milne Edwards, 1875)	+	+				
<i>Munida sanctipauli</i> Henderson, 1885		+				
<i>Pachygrapsus corrugatus</i> (von Martens, 1872)	+	+	+			
<i>Panulirus argus</i> (Latreille, 1804)	+	+				
<i>Panulirus echinatus</i> Smith, 1869		+	+	+	+	
<i>Plagusia depressa</i> (Fabricius, 1775)	+	+	+	+	+	+
<i>Sergestes edwardsii</i> Kroyer, 1855		+				
<i>Stenorhynchus spinifer</i> Miers, 1886		+				
<i>Synalpheus fritzmuelleri</i> Coutière, 1909	+	+	+	+		
<i>Tesseropora atlantica</i> Newman & Ross, 1977		+				
<i>Xanthodius denticulatus</i> (White, 1848)	+	+	+			+
Deep Sea						
<i>Chaceon gordonae</i> (Ingle, 1985)		+	+	+		+
<i>Bathynectes longispina</i> Stimpson, 1871	+	+				
<i>Dromia bollorei</i> Forest, 1974		+	+			+
<i>Homola minima</i> Guinot & Richer de Forbes, 1995	+	+				
<i>Nematocarcinus gracilipes</i> Filhol, 1884		+			+	+
<i>Plesionika martia</i> (A. Milne-Edwards, 1883)		+				+

The zoogeographic aspects among the deep-sea species of this study indicate that the SPSPA is a transition zone between the American and African continents. For example, *Homola minima* is a species previously found exclusively on western Atlantic (Manning & Holthuis, 1981), while *Bathynectes longispina* and *Dromia bollorei* are widespread in tropical Atlantic, with SPSPA being the western and eastern limit of their range, respectively (Manning & Holthuis, 1981; Tavares 2003; Torres *et al.*, 2006). *N. gracilipes* had been already described for the Mid-Atlantic Ridge, from the Azores to Brazil (Cardoso & Burukovsky, 2014); and *Plesionika martia* had been reported for both the Atlantic and the Pacific oceans, but this is the first record of the species for the SPSPA.

The presence of American species, West African species, species of the tropical islands of the central Atlantic (Ascension and St Helena) and Mid-Atlantic Ridge (Azores and Cape Verde Archipelago) demonstrates the biological importance and strategic location of SPSPA. It also shows the zoogeographical influence of the South Equatorial Current, flowing at the surface, in the east-west direction, and of the Equatorial Undercurrent, flowing under the surface (60–100m) in the opposite direction. These opposing currents allow the colonization of species from both sides of the Atlantic Ocean, that have the ability to adapt to the ecosystem of the SPSPA (Bowen, 1966;

Edwards & Lubbock, 1983; Stramma & Schott, 1999), to settle there. The findings of brachyuran and caridean larvae, in relatively high abundance, around the SPSPA confirm the hypothesis that it does support a complex community of crabs and shrimps (Díaz *et al.*, 2009; Koettker *et al.*, 2010; Brandão *et al.*, 2013).

The dominant species (*C. gordonae*) was found in all depths, except between 100 and 200 m. *B. longispina*, in turn, was found only in shallower layers, between 100 and 300 m, while *D. bollorei* and *H. minima* were restricted to the range of 400 to 600 m. *N. gracilipes* was collected between 300 and 700 m, and a just a single specie of *P. martia* was caught at 340 m depth. All this information is crucial to understand the vertical distribution of these species and the factors that control it, such as light, food availability and reproduction (Pezzuto & Sant'Ana, 2009; Ferreira *et al.*, 2016). The depth distribution of these species in the present study also corroborates with studies in other regions (Manning & Holthuis, 1981; Afonso-Dias *et al.*, 2008; Torres *et al.*, 2006; Cardoso & Burukovsky, 2014; Tavares & Lamaitre, 2014).

These results evidence the still acute lack of basic information and taxonomic studies on benthic fauna in these environments (Martins *et al.*, 2016; Anker *et al.*, 2016; Anker *et al.*, 2007), particularly with regard to deep-sea decapod crustaceans. Recently, Tavares *et al.* (2017) conducted shallow water surveys on Trindade and Martin Vaz Archipelago and found ten new records of decapod crustaceans there, highlighting the need for studies on the faunistic similarities between the tropical islands of the South Atlantic Ocean, as in northeastern Brazil (Archipelago Of São Pedro and São Paulo, Atol das Rocas and Fernando de Noronha). Therefore, further efforts should be undertaken in the coming years, using different sampling methods, to access and describe this still largely unknown biodiversity.

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